数据架构规划（注释英文版）

DF1 Framework

代码块

```
dataset/
├── meta/
│   ├── task/
│   │   ├── sop/                          # Task description, steps and execution time; task_steps not split yet
│   │   └── illustrations/                # Task reference materials; can be omitted if empty
│   ├── scene/                            # Optional: scene identity and necessary background for future scene association
│   └── annotation/                       # Planned: future descriptive annotations
│       └── language/
│           ├── self_speech/              # Future self-speech transcription, timestamps and audio references
│           ├── external_speech/          # Future external speech transcription, speakers and timestamps
│           └── prompt/                   # Future actual prompt content, sources and timestamps
├── observation/
│   ├── timeline/                         # Main timeline and reference sources, following original design
│   ├── image/                            # Video, timestamps, and intrinsic/extrinsic parameters of each camera
│   │   ├── rgb_wrist_left/
│   │   ├── rgb_wrist_right/
│   │   ├── rgb_wrist_left_up/            # Planned: future three-finger left thumb-side upper camera
│   │   ├── rgb_wrist_left_down/          # Planned: future three-finger left thumb-side lower camera
│   │   ├── rgb_wrist_right_up/           # Planned: future three-finger right thumb-side upper camera
│   │   ├── rgb_wrist_right_down/         # Planned: future three-finger right thumb-side lower camera
│   │   ├── rgb_ego_center_left/
│   │   ├── rgb_ego_center_right/
```

```
23  │   │   ├── rgb_ego_side_left/
24  │   │   ├── rgb_ego_side_right/
25  │   │   ├── ir_ego_left/
26  │   │   └── ir_ego_right/
27  │   ├── audio/                          # Optional planned: future raw audio
    recording
28  │   ├── state/
29  │   │   ├── head/                       # Optional planned: future head-
    mounted device online SLAM localization results
30  │   │   │   └── pose/                   # Device pose in the SLAM reference
    frame
31  │   │   ├── body/                       # Planned: future full-body motion
    capture / pose estimation results
32  │   │   │   ├── root/                   # Root node position, orientation
    and overall movement trajectory
33  │   │   │   ├── joints/                 # Full-body joint poses, including
    head
34  │   │   │   └── motion/                 # Future: save velocity, angular
    velocity and other motion quantities as needed
35  │   │   ├── left_hand/
36  │   │   │   ├── encoder/                # Raw encoder angles
37  │   │   │   ├── handpose/               # Optional planned: future online
    left-hand pose estimation
38  │   │   │   └── tactile/                # Tactile data
39  │   │   └── right_hand/
40  │   │       ├── encoder/                # Raw encoder angles
41  │   │       ├── handpose/               # Optional planned: future online
    right-hand pose estimation
42  │   │       └── tactile/                # Tactile data
43  │   └── env/
44  │       └── environment_map/            # Planned: future mapping-derived
    maps or scene reconstruction
45  └── accessories/                        # External auxiliary input for pose
    and tactile processing
46      ├── imu/
47      │   ├── imu_ego/                    # Migrated from existing RING IMU,
    original attributes preserved
48      │   ├── imu_wrist_left/             # Planned: future left wrist IMU
    input
49      │   └── imu_wrist_right/            # Planned: future right wrist IMU
    input
50      ├── emg/                            # Planned: future left and right EMG
    acquisition input
51      │   ├── emg_left/
52      │   └── emg_right/
```

```
53      └── mocap/                          # Planned, optional: future motion capture raw input
```

DF2 Framework

代码块

```
1   /
2   └── dataset/
3       ├── meta/
4       │   ├── task/
5       │   │   ├── sop/                    # Task description, operation steps and execution time
6       │   │   └── illustrations/          # Task reference materials; can be omitted if empty
7       │   │
8       │   ├── scene/                      # [Optional] scene identity and necessary background for future scene association
9       │   │
10      │   └── annotation/                 # [Planned] future descriptive annotations
11      │       └── language/
12      │           ├── self_speech/        # Self-speech transcription, timestamps and audio references
13      │           ├── external_speech/    # External speech transcription, speakers and timestamps
14      │           └── prompt/             # Actual prompt content
15      │
16      ├── observation/
17      │   ├── timeline/                   # Main timeline
18      │   │
19      │   ├── image/
20      │   │   ├── rgb_wrist_left/
21      │   │   ├── rgb_wrist_right/
22      │   │   ├── rgb_wrist_left_up/
23      │   │   ├── rgb_wrist_left_down/
24      │   │   ├── rgb_wrist_right_up/
25      │   │   ├── rgb_wrist_right_down/
26      │   │   ├── rgb_ego_center_left/
27      │   │   ├── rgb_ego_center_right/
28      │   │   ├── rgb_ego_side_left/
29      │   │   ├── rgb_ego_side_right/
30      │   │   ├── ir_ego_left/
31      │   │   └── ir_ego_right/
```

```
32      │   │
33      │   ├── audio/                          # [Optional planned] future
    raw audio recording
34      │   │
35      │   ├── state/
36      │   │   ├── body/                       # [Planned] future full-body
    motion capture or pose estimation results
37      │   │   │   ├── root/                   # Root node position,
    orientation and overall movement trajectory
38      │   │   │   ├── joints/                 # Full-body joint poses,
    including head
39      │   │   │   └── motion/                 # Save velocity, angular
    velocity and other motion quantities as needed
40      │   │   │
41      │   │   ├── left_hand/
42      │   │   │   ├── handpose/               # Left wrist pose
43      │   │   │   ├── joints/                 # Left hand joint states
44      │   │   │   └── tactile/                # Left hand tactile data
45      │   │   │
46      │   │   └── right_hand/
47      │   │       ├── handpose/               # Right wrist pose
48      │   │       ├── joints/                 # Right hand joint states
49      │   │       └── tactile/                # Right hand tactile data
50      │   │
51      │   └── env/
52      │       ├── objects/
53      │       │   └── obj_1/                  # Object pose
54      │       └── environment_map/            # [Planned] save environment
    map or scene reconstruction results after mapping
55      │
56      └── accessories/                        # [Planned] future external
    auxiliary input
57          ├── imu/
58          │   ├── imu_ego/                    # Saved when Ego IMU is
    integrated in the future
59          │   ├── imu_wrist_left/             # Future left wrist IMU input
60          │   └── imu_wrist_right/            # Future right wrist IMU input
61          │
62          ├── emg/                            # Future left and right EMG
    acquisition input
63          │   ├── emg_left/
64          │   └── emg_right/
65          │
66          └── mocap/                          # Optional: future motion
    capture raw input
```